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AI-infused knowledge and green intellectual capital: pathways to spur accounting performance drawn from RBV-KBV model and sustainability culture

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ABSTRACT

AI-infused knowledge (AIK) systems have the potential to transform the way firms operate, providing dynamics for optimizing processes, enhancing decision-making, and fostering innovation. This shift can play a vital role in service-oriented contexts, like accounting firms. In this study, we aim to explore how AIK can drive green intellectual capital (GIC) aspects, spurring accounting firms' performance related to environmental and sustainable practices. Drawn from the resource-based view (RBV) and knowledge-based view (KBV), sustainability culture (SUC) is integrated into the model as a moderator variable to provide an in-depth understanding of this matter. The study adopted a quantitative approach, data was collected through an online survey distributed to accounting professionals. The analysis shows interesting outcomes: AIK positively and significantly affected the internal dimensions of GIC, green human capital (GHC) and green structural capital (GSC) within accounting firms, while green rational capital (GRC) had no effect. In turn, these two internal dimensions, GHC and GSC, had a significant impact on accounting performance across various aspects, including efficiency, accuracy, compliance with environmental standards, innovation in reporting, and timeliness. Intriguingly, SUC was found to significantly moderate the association between GIC dimensions and accounting firm performance. The results of this research offer actionable insights for stakeholders and decision-makers, including accounting firms, clients, and the community at large. Accounting firms are highly recommended to adopt AI-infused platforms to foster GIC dynamics, specifically GHC and GSC. AI-infused knowledge tools have the potential to improve efficiency, accuracy, innovation in reporting, compliance with sustainability standards and timeliness, thus leading to better overall performance. Although service-oriented contexts, like accounting, have the potential to substantially benefit from GIC practices, research in such a significant discipline has been remarkably scarce.

1. Introduction

In today's rapidly evolving business landscape, artificial intelligence (AI) technologies are transforming the way organizations operate, offering tools for optimizing processes, enhancing decision-making, and fostering innovation (Enholm et al., 2022; Loureiro et al., 2021). This shift has significant implications for the accounting industry (Han et al., 2023; Norzelan et al., 2024), where AI-infused knowledge (AIK) systems are enabling firms to streamline tasks, improve accuracy in reporting, and enhance overall performance (Anh et al., 2024). These systems, which encompass advanced data processing, automated analytics, and decision-support capabilities, are seen as critical assets in maintaining competitive advantage (Ahmad et al., 2023; Norzelan et al., 2024). However, the broader implications of integrating AI into intellectual capital (IC), particularly green IC (GIC), remain an underexplored area of research.

Firms across various industries, including accounting firms, have been forced to rethink their business strategies in recent years due to the growing importance of sustainability in the corporate world (Hsiao et al., 2022; Rodrigue et al., 2023). Regulations and the general movement toward environmental responsibility are putting increasing pressure on firms, particularly in the accounting industry, to adopt

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sustainable practices. In order to meet the demands of stakeholders and environmental goals, accounting firms that have traditionally prioritized financial reporting and compliance must now integrate sustainability-related practices (Channuntapipat et al., 2020). In this transition, GIC, consisting of green human capital (GHC), green structural capital (GSC), and green relational capital (GRC), has been identified as pivotal in driving sustainable development in service-oriented contexts. GIC represents the knowledge, structures, and relationships that support sustainability within organizations (Umar et al., 2024; Shehzad al., 2023; Yusliza et al., 2020). Unlike other service industries, the accounting industry functions in a strict regulatory framework combined with extensive datasets needed for accurate work that demands both precision and compliance and transparency (Bin-Nashwan et al., 2025). AIK delivers advanced financial analysis automation together with accurate audits and better risk assessment and regulatory compliance through automated processes (Wolters Kluwer, 2024). AI enables sustainable financial reporting as well as green investment advisory and ethical tax planning services for accounting firms to become strategic sustainability partners (Li et al., 2023; Nishant et al., 2020). By strengthening green human capital, structural capital, and relational capital, AIK not only enhances operational efficiency but also fosters long-term competitiveness in the accounting profession. Despite this, there remains a noticeable gap in understanding how AIK interacts with different components of GIC, particularly the varying influence it has on GHC, GSC, and GRC (Almansour, 2024). AI technologies may enhance human and structural capital by fostering green skills and embedding sustainable practices in organizational systems, the relational dimension—built on stakeholder engagement and trust—might not benefit as directly from AI interventions, as some prior studies argued (Asiaei et al., 2022). The objective of our research is to explore how AIK can drive green intellectual capital (GIC) aspects, spurring accounting firms' performance related to environmental and sustainable practices, as well as the moderating role of sustainability culture in the proposed model.

Theoretically, while the resource-based view (RBV, Barney, 1991) and knowledge-based view (KBV, Grant, 1996) offer a foundation for understanding how firms leverage intangible resources like GIC to spur optimal firm performance (Al Issa et al., 2023), research on the integration of AIK systems in enhancing these green capabilities, particularly in the accounting sector, is very scarce. AIK systems, when aligned with GIC, can be seen as a resource that enhances an organization's ability to adapt to environmental challenges and regulatory requirements (Anh et al., 2024; Saxena, 2024; Wang & Juo, 2021). Through AI-powered knowledge the operational efficiency of a firm increases through automated complex tasks and enhanced decision systems and tight regulatory adherence. The AI-based improvements help GIC development through its three core dimensions including human capital and structural capital and relational capital. Accounting firms gain improved performance abilities through increased development of GIC elements. Additionally, the role of sustainability culture as a moderating factor in these relationships has not been fully investigated. The collective values and environmental commitments and organizational practices toward sustainable development known as sustainability culture. It has the potential to strengthen GIC performance impact through an environment dedicated to sustainable development goals (Marshall et al., 2015).

The novelty of our research can be justified by the following.

(1) Literature has mainly paid attention to the traditional aspects of IC and firm performance (Bayraktaroglu et al., 2019; Truong et al., 2024; Xu & Li, 2022), GIC and firm performance (Hoang Thanh & Truong Cong, 2024; Shehzad et al., 2023; Umar et al., 2024), and the role of IC in firms' competitive advantage (Ahmad et al., 2023; Kamukama & Sulait, 2017). However, there is a dearth of research exploring the crucial impact of AIK on GIC, and thus fostering firms' performance. Therefore, our study is likely the first of its kind to delve empirically into the interplay between AIK on GIC.
(2) The GIC—firm performance relationship was examined in the literature to a reasonable extent, it is not overly saturated compared to traditional IC. Prior studies have investigated the effect of GIC on performance in certain domains, such as manufacturing (Umar et al., 2024; Shehzad al., 2023; Yusliza et al., 2020; Mansoor et al., 2019), hospitality (Elzek et al., 2024; Haldorai et al., 2022; Shahbaz et al., 2024), high-tech industry (Wang & Juo, 2021), SMEs (Khanlarov et al., 2020, pp. 100–109), healthcare (Zahlan et al., 2023), energy (Liu et al., 2021), education (Ahmed et al., 2023), and waste recycling (Haddad et al., 2024). Yet, although service-oriented contexts, like accounting, also have the potential to benefit from GIC practices, research in such a significant discipline has been remarkably scarce. Green strategies adopted in accounting firms could spur environmental reporting, improving stakeholder engagement around sustainability and optimizing internal processes to be eco-friendlier. Hence, our study enriches the literature by examining the interplay of AIK and GIC within accounting firms and how this can spur their performance.
(3) Our study also contributes to the body of knowledge by incorporating sustainability culture as a moderator, strengthening the association between GIC and accounting firm performance.
(4) The integrated model of RBV-KBV could provide a viable and comprehensive framework for AI-infused-GIC and firm performance. This model could serve as a catalyst for further literature in the field.
(5) From a practical perspective, we address a pressing issue for accounting firms, as they face increasing expectations to balance financial performance with sustainability. Compliance with environmental regulations, innovation in green reporting practices, and the efficient use of resources are becoming key performance indicators. The integration of AI technologies offers a pathway for firms to enhance their GIC, but the success of these technologies' hinges on the presence of a strong sustainability culture. Understanding these dynamics can offer valuable insights for accounting firms aiming to enhance their sustainability practices while improving their operational efficiency and competitiveness.

This research also offers actionable insights for stakeholders and decision-makers, including accounting firms, clients, and the community at large. Accounting firms are highly recommended to adopt AI-infused platforms to foster GIC dynamics, specifically GHC and GSC. AI-infused knowledge tools have the potential to improve efficiency, accuracy, innovation in reporting, compliance with sustainability standards and timeliness, thus leading to better overall performance. Hence, our research strives to answer the following questions.

- How does AIK drive GIC components within accounting firms?
- Do GIC dimensions spur accounting firm performance?
- Does sustainability culture moderate the nexus between GIC and firm performance?

The work unfolds across Six Main Parts. Following the introduction in Part 1, literature review and hypothesis development are presented in Part 2. Research methodology including data collection processes and analysis are outlined in Part 3. In Part 4, research results are presented and addressed in light of the theoretical framework, as well as research implications for theory and policy. Part 5 wraps up the journey by distilling the essence of findings, policy recommendations, and potential directions for future research.

2. Literature review

2.1. Theoretical basis: RBV-KBV framework

The RBV, popularised by Barney (1991), posits that firms can achieve a competitive advantage by possessing and utilizing valuable, rare, inimitable, and non-substitutable (VRIN) resources. The collection of resources consists of physical assets with human talents and organizational system management and mental knowledge. RBV explains performance differences between firms through resource heterogeneity and mobility limitations because some organizations hold distinct resources which competition finds difficult to duplicate (Arend and Lévesque, 2010; Kozlenkova et al., 2014). Accounting firms which successfully utilize their GIC including sustainability expertise together with environmentally friendly technologies and stakeholder relationships obtain superior performance as well as increased market competitiveness (Ahmad et al., 2023; Kamukama & Sulait, 2017). The advanced sustainability-related resources found in certain organizations enhance their reporting practices by delivering better efficiency together with compliance and innovative capabilities. The RBV coupled with sustainability efforts enables organizations to reach operational excellence and satisfy regulatory and societal standards according to Arda et al. (2023) and Samadhiya and Agrawal (2024). Unique green resources enable organizations to boost performance through compliance and innovative practices in addition to developing beneficial reputations (Barney, 2001; Hart, 1995; Khanra et al., 2022).

The KBV extends the RBV by emphasizing that knowledge is the most critical resource in modern organizations (Cuthbertson & Furseth, 2022). According to Grant (1996), firms are repositories of knowledge, and their success depends on how effectively they create, store, and apply knowledge. This view focuses on how organizations manage intellectual capital, particularly through human capital, relational capital, and structural capital, to foster innovation and performance (Low & Ho, 2016; Zheng et al., 2010; Eisenhardt & Santos, 2002). For accounting firms, KBV underlines the importance of leveraging knowledge-driven processes, particularly in sustainability practices (Azmi et al., 2024). Firms that manage their green human capital (employee skills in sustainability), green structural capital (systems and processes that support sustainability), and green relational capital (stakeholder relationships focused on environmental concerns) are more likely to achieve superior accounting performance (Asiaei et al., 2022). Integrating AI-infused knowledge systems can further enhance these knowledge capabilities, allowing firms to generate insights, optimize processes, and increase compliance and reporting efficiency (Nonaka, 2009; Teece, 2007). By combining insights from both RBV and KBV, we can argue that firms that leverage both their unique green resources and advanced knowledge capabilities are better equipped to achieve superior accounting performance, especially in areas such as compliance, accuracy, and innovation in reporting.

2.2. AI-infused knowledge and green intellectual capital

AI-infused knowledge (AIK) plays a transformative role across various industries, enhancing sustainability and improving firm performance. In manufacturing, AIK optimizes production processes by reducing waste and improving energy efficiency (Jing & Zhang, 2024). In healthcare, AIK supports diagnosis and enhances operational efficiency, leading to better patient outcomes and reduced costs (Dutt, 2020). In finance, AIK improves risk assessment, automates financial analysis, and ensures regulatory compliance (Milana & Ashta, 2021). AIK enables the retail industry to enhance demand forecasting as well as inventory management and personalized customer experiences which saves resources by utilizing them more efficiently (Oosthuizen et al., 2021). Digitized metrological knowledge through AIK helps farmers practice precise agriculture which optimizes water conservation and minimizes chemical product waste for sustainable food management (Javaid et al., 2023). AIK demonstrates usefulness for improving educational environments and management systems in the educational sector according to Bouteraa et al. (2024) and Sadallah et al. (2025). The use cases outline AIK's wide applicability in various sectors for supporting sustainable operations as well as process optimization (Cubric, 2020).

AIK systems have emerged as fundamental organizational tools to upgrade GHC performance according to Almansour (2024) and Kusi-Sarpong, Mubarik, Khan, Brown, and Mubarak (2022). Firms that utilize AI technologies can create training programs which deliver customized sustainability development experiences to employees. Such employee training builds essential green initiative competence which leads to sustainability culture development according to AL-Khatib and Shuhaiber (2022) and Gashenko et al. (2020). Employees' learning patterns undergo AI analysis so that customized materials become available which follow sustainability-oriented organizational objectives (Cicerone et al., 2023). Through targeted training programs organizations can boost environmental consciousness and create possibilities for their employees to implement innovative practices which drive sustainable development in the company (Abdelfattah et al., 2024).

GSC includes all operations' systems and processes together with infrastructure that supports sustainable operations (Hoang Thanh & Truong Cong, 2024; Shehzad et al., 2023). Knowledge augmented by AI capabilities boosts GSC through the optimal management of operations and resources (Ruzza et al., 2020). AI evaluation of extensive records enables managers to locate operational inefficiencies that lead to better resource decisions for waste reduction (Li et al., 2023; Nishant et al., 2020). Organizations using structural capital with AI-powered systems can develop superior environmental management systems to precisely monitor and present sustainable initiative progress (Cheng et al., 2024). The connected operations prove beneficial for operational effectiveness and sustainability goals by improving organizational performance.

The interplay between AIK systems and GRC presents a more nuanced picture. The sustainability practices through GRC rely on collaborative networks between stakeholders for their achievement (Wu & Yu, 2024). The implementation of AIK systems within accounting firms generates substantial GRC improvement through stronger connections with environmentally oriented stakeholders and both clients and legal bodies. The external social networks through which a firm supports its sustainability initiatives are known as GRC. The implementation of AI tools including data analytics and environmental reporting automation and advanced client management systems through accounting firms lets them serve sustainability-conscious clients interested in corporate social responsibility (CSR) (Bharadwaj et al., 2013; Fukuda-Parr & Gibbons, 2021 Fukuda-Parr & Gibbons, 2021). AI improves sustainability data reporting accuracy through automated environmental process management which enhances stakeholders' trust about environmental concerns and client faith in accounting firms' sustainability practices (Lourenço, Branco, Curto, & Eugénio, 2012). AI systems evaluate extensive information about clients to find sustainability possibilities and risks which enables accounting firms to give advanced environmentally-oriented counsel that creates strong partnerships through shared sustainability aims. The implementation of AI systems helps accounting firms develop strong relations with sustainability-related regulatory bodies and industry organizations. The integration of AI technology with blockchain features delivers improved data visibility allowing firms to validate their adherence to growing sustainable regulations (Grewal, Riedl, & Serafeim, 2019). AI can also assist in monitoring clients' adherence to environmental standards and integrating sustainability metrics into financial performance assessments. As accounting firms increasingly incorporate sustainability into their service offerings, AI helps build and maintain GIC by aligning the firm's capabilities with clients' environmental goals and expectations (Goralski, 2020; Goralski & Tan, 2020). By leveraging AI to support sustainability initiatives, we can assume that accounting firms can create deeper, more meaningful connections with stakeholders

committed to green practices.

H1. AI-infused knowledge is positively associated with GHC.

H2. AI-infused knowledge is positively associated with GSC.

H3. AI-infused knowledge is positively associated with GRC.

2.3. GIC and accounting performance

Literature has mainly paid attention to the traditional aspects of intellectual capital and firm performance (Bayraktaroglu et al., 2019; Truong et al., 2024; Xu & Li, 2022), the role of intellectual capital in firms' competitive advantage (Ahmad et al., 2023; Kamukama & Sulait, 2017), green intellectual capital and firm performance (Hoang Thanh & Truong Cong, 2024; Shehzad et al., 2023; Umar et al., 2024). As the first dimension of GIC, GHC refers to employees' skills, knowledge, and competencies regarding environmental sustainability. Employees equipped with sustainability-related skills contribute to improved decision-making processes and enhance the firm's capability to meet regulatory requirements. Additionally, training programms that focus on green practices not only increase employee awareness but also foster a culture of sustainability within firms, leading to better financial outcomes (Munawar et al., 2022; Zhou et al., 2023). Previous studies indicated that GHC contributes to enhanced firm performance by fostering a culture of environmental awareness and accountability among employees (Hoang Thanh & Truong Cong, 2024; Shehzad et al., 2023). The ongoing development of GHC by firms demonstrates their environmental responsibility at market levels and enables better interactions concerning stakeholders such as clients and investors and regulatory bodies. Accounting firms can strengthen multiple performance aspects through the implementation of GHC because it enhances operational efficiency as well as reporting accuracy and innovation and delivers prompt service while achieving environmental standards compliance (Abernathy et al., 2014; Chan & Vasarhelyi, 2018; DeFond & Zhang, 2014; Mancini et al., 2017). Positive sustainability reputation creates trust while boosting credibility which serves as essential drivers for enduring business success (Pham et al., 2020).

H4. GHC positively influences accounting firms' performance.

GSC systems in companies embody environmental programs that require process establishment and structural development and continuous integration maintenance of all participating elements. Prior research validated that firms possessing exceptional GSC demonstrate enhanced accounting performance based on Li et al. (2023) and Yusliza et al. (2020). When organizations implement thorough environmental management systems they gain enhanced sustainability monitoring abilities leading to workflow enhancement that minimizes operational costs. Accounting performance increases through GSC because organizations manage their resources exactly and decrease waste spending (Umar et al., 2024). Financial results improve when stakeholders trust GSC due to its transparent practices (Yu et al., 2021). GSC enables accounting firms to deploy systems of organizational structure with technology frameworks and sustainability policies to establish environmental sustainability in their operations. Research has established that companies dedicating attention to GSC accomplish enhanced performance related to regulatory acceptance and innovative sustainable reporting practices and operational success (Shehzad al., 2023; Ullah et al., 2022; Ali et al., 2022; Yusliza et al., 2020). Accounting firms establish readiness for legal and stakeholder compliance needs through effective implementation of environmental management systems which links compliance tasks with sustainability metrics. Environmental rule compliance improves and sustainability disclosures become stronger when firms establish proper alignment according to research. Understanding and accountability development through transparency creates trustworthy relationships with dedicated customers.

H5. GSC positively influences accounting firms' performance.

GRC is about relationships a firm maintains with its stakeholders' concerning sustainability. The literature indicates that strong GRC can significantly enhance firms' performance by fostering collaboration and knowledge sharing among stakeholders (Nirino et al., 2022; Yu et al., 2021). Companies that invest in green rational capital by actively engaging with their stakeholders, e.g., clients, suppliers, and community organizations, on sustainability issues often experience improved brand reputation and client loyalty. This can also help differentiate themselves in a competitive market and ultimately enhance performance (Yadiati et al., 2019). By establishing themselves as trusted advisors on sustainability reporting and environmental compliance, accounting firms with high GRC can enhance their reputation and strengthen client relationships, ultimately leading to improved financial performance and client retention. Green rationality drives innovation and value creation within accounting firms, which translates into higher performance (Ali et al., 2022; Ullah et al., 2022; Umar et al., 2024; Shehzad al., 2023; Yusliza et al., 2020). When firms engage deeply with stakeholders who value sustainability, they gain insights into emerging sustainability trends, standards, and client needs (Zhou et al., 2023). This ongoing collaboration encourages accounting firms to innovate in areas such as green auditing, sustainability reporting, and compliance consulting services. However, the impact of GRC on accounting performance is complex, as it may also depend on the firm's ability to leverage these relationships for innovative solutions and sustainable practices (Juo & Wang, 2022). Thus, firms need GRC to achieve better financial outcomes from their sustainable initiatives.

H6. GRC positively influences accounting firms' performance.

2.4. Sustainability culture (SUC) as a moderator

Sustainability culture (SUC) embodies a set of shared values, practices, and beliefs to prioritise sustainable practices and environmental responsibilities within companies and among their employees (Al-Emran, 2023; Al-Emran & Griffy-Brown, 2023; Marshall et al., 2015). Companies implement sustainability initiatives and their overall performance results from their cultural orientation (Galpin et al., 2015; Haladu & Bin-Nashwan, 2024; Haladu et al., 2025a, 2025b; Haladu & Bin-Nashwan, 2022). The research demonstrates that businesses with robust sustainability culture frameworks achieve better financial outcomes from their global information centers (GIC) (Schönborn et al., 2019; Siyal et al., 2022). Organizations adopting a sustainable strategic approach prioritise sustainability in their decisions which produces more efficient reporting processes.

Likewise, SUC can play a critical role in bridging the gap between GIC components and organizational performance outcomes. Although the existing literature that explored SUC appears very limited, Schönborn et al. (2019) suggested that a strong sustainability culture can amplify the positive effects of GHC on innovation and operational effectiveness, as employees are more likely to engage in environmentally friendly practices when supported by an organizational culture that values sustainability (Ketprapakorn & Kantabutra, 2022; Salvioni & Almici, 2020). Scholars, such as Yong et al. (2019), have shown that employees' skills and motivations around environmental initiatives have a greater effect on performance when backed by a sustainability-driven organisational culture, reinforcing environmental goals, and leading to improved productivity, innovation, and compliance. On the other hand, in firms with a poor sustainability culture, GHC benefits may not materialise as intended, resulting in missed opportunities for improving performance.

Firms with high sustainability cultural values enhance the impact of eco-oriented policies through daily operation integration while increasing strategic decision effectiveness (Salvioni & Almici, 2020). According to past studies (Liu et al., 2021; Schönborn et al., 2019), firms with an established sustainability culture not only improve compliance and reporting accuracy but also increase operational efficiency through

sustainable resource use. This alignment strengthens organisational capabilities and optimizes performance metrics, especially in areas like cost savings and resource management. Likewise, GRC finds value through SUC because the approach helps organizations uphold enduring relationships with stakeholders through mutual environmental commitments (Siyal et al., 2022). When SUC supports GRC, firms tend to experience greater stakeholder loyalty and trust, leading to enhanced market reputation and brand differentiation (Yu et al., 2021). As such, firms with a strong culture of sustainability can better leverage their relational capital, translating stakeholder goodwill into competitive advantages and improved operational and financial outcomes. In the present study, we assume that sustainability-oriented culture serves as a crucial moderator that can transform GIC into tangible performance benefits for service-oriented contexts like accounting firms. Fig. 1 displays our proposed research model.

H7. SUC moderates the relationship between GHC and accounting firms' performance.

H8. SUC moderates the relationship between GSC and accounting firms' performance.

H9. SUC moderates the relationship between GRC and accounting firms' performance.

3. Research methods

3.1. Sampling and data collection

The online survey remains popular because it offers researchers benefits which include formulating research models and hypotheses together with affordability and direct access to the audience as well as fast response options (Fricker & Schonlau, 2002). The researchers have selected an online survey as the optimal research method for their study on AI tools (Bin-Nashwan, Sadallah, & Bouteraa, 2023). Our research determines the ways ChatGPT integration impacts GIC operations to achieve better accounting outcomes from an accounting professional viewpoint. The selection of accounting professionals for this study was attributed to the fact that they actively work with extensive data while managing compliance needs and making decisions where AIK significantly affects their activities (Bin-Nashwan et al., 2025). Professional accountants supply essential knowledge about AIK's potential to increase operational effectiveness as well as sustainability for this industry sector. The findings can produce beneficial implications for different service-based sectors which possess similar traits including strict regulatory oversight and data-driven business approaches. The online survey took place from December 21, 2023 to January 7, 2024. Over 5000 invitations via online platforms were sent to Chinese panelists to participate in the study. The sample was chosen randomly and fulfilled specific criteria, including accounting firms, and to be responded to by accounting professionals, such as chief financial officers (CFOs), audit managers, senior accountants, or senior auditors. The chosen participants can provide comprehensive practical, technical, and strategic insights. To effectively filter participants and ensure a qualified sample that aligns with the research objectives (Short et al., 2002), we included a screening question: "Has your firm already adopted AI in accounting services?" This question was designed to ensure that participants have direct experience with AI implementation, as their insights are essential to understanding the impact of AIK on accounting firm performance. By using this criterion, we focused on firms where AI adoption is actively influencing operations.

The study participants received a detailed description of the study's objectives and scope together with confirmation their responses would remain private and would only be used for academic investigations. The survey questions were arranged differently to prevent possible bias from influencing responses. The research eliminated participants who finished each question within 5 s or less because such rapid answers point toward unreliable data in digital surveys (Bin-Nashwan & Muneeza, 2023). automated systems removed participants who finished the survey questions in unusually short amounts of time or repeated answers without hesitation as they likely did not fully consider their responses. The survey included 461 respondents who managed to pass the screening criterion among all who took the assessment. An analysis was performed using 433 valid responses because the remaining data underwent an outlier exclusion process.

3.2. Measurements

The researchers used 44 indicators in an online survey to assess 6 constructs which were AI-infused knowledge and green human capital, green structural capital, green rational capital, sustainability culture and accounting performance. The study used modified items derived from established literature scales which met the requirements of AI-infused knowledge-GIC integration in accounting. Table 1 shows details about the construct definitions and sources for measurements.

The survey implemented a seven-point Likert scale to evaluate questions which led to accomplished discriminant and reliability validity as per Preston and Colman (2000). The first English version of the survey received translation from three expert bilingual personnel into Chinese after which we employed back-translation into English. An English back-translation of the Chinese questionnaire was performed using the approach mentioned in Douglas and Craig (2007).

Three specialists from the fields of AI technologies alongside accounting reviewed the survey for content validation purposes. Three accounting experts checked the content relevance of the survey, and

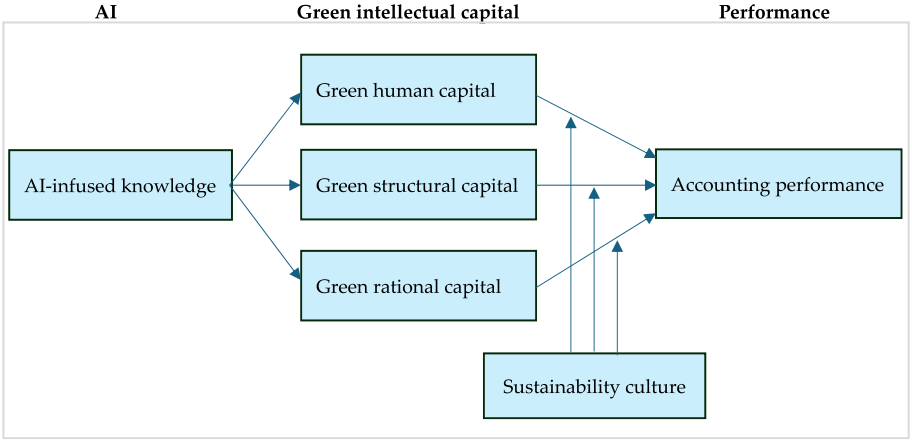


Fig. 1. Research model.

Table 1
Measurements of research constructs.

Construct	Conceptualisation/Operationalisation	Source
AI-infused knowledge (AIK)	AI technologies (e.g., machine learning, natural language processing, and data analytics) designed to manage, analyse and disseminate knowledge within accounting firms. Such systems can facilitate real-time data analysis, automate routine tasks and improve the precision of financial reporting and compliance efforts. AIK improved by learning from data, enabling accounting firms to manage complex environmental data, optimize green process and foster sustainable innovation. AIK was measured by five items as follows: AIK1: Our firm effectively utilizes AI-driven insights to enhance decision-making and strategic planning. AIK2: AI-powered tools in our firm significantly improve the accuracy and efficiency of financial reporting. AIK3: Employees in our firm are well-trained to integrate AI technologies into their daily accounting practices. AIK4: Our firm leverages AI systems to optimize sustainability-related data collection and analysis. AIK5: AIK helps our firm develop innovative solutions for improving environmental and sustainability performance.	Laupichler et al. (2023)
Green human capital (GHC)	Professionals' knowledge, skills, expertise, and competences within accounting firms in sustainability reporting, carbon accounting, and environmental regulations. GHC was measured using four items as follows: GHC1: The contribution of environmental protection by employees in our firm is better than our major competitors. GHC2: Employee competence with respect to environmental protection in our firm is better than that of our major competitors. GHC3: Our managers fully support our employees in achieving their goals with respect to environmental protection. GHC4: Employees in our firm are actively engaged in sustainability training and environmental protection initiatives.	Wang and Juo (2021)
Green structural capital (GSC)	Organisational structures and processes within accounting firms that support their sustainable practices and innovations. Six items were used to measure GSC as follows: GSC1: The management system for environmental protection in our firm is superior to that of our major competitors. GSC2: Investments in environmental protection facilities in our firm are more than those of our major competitors. GSC3: Competence in developing green products in our firm is better than that of our major competitors. GSC4: The overall operational processes for environmental protection in our firm work smoothly. GSC5: The knowledge management system for environmental management in our firm is favorable for the accumulation of knowledge related to environmental management. GSC6: Our firm's internal communication and collaboration systems effectively support environmental protection initiatives.	Wang and Juo (2021)
Green rational capital (GRC)	The ability of accounting firms to interact with various external stakeholders, e.g., clients, auditors, regulators, competitors, investors, and suppliers, to foster collaboration and commitment to sustainability. GRC was measured using seven items, as follows: GRC1: Our firm has established strong partnerships with external stakeholders for promoting environmental protection. GRC2: Our firm actively shares environmental protection information and practices with external stakeholders. GRC3: Our customers recognize and support our efforts toward environmental sustainability. GRC4: Our firm frequently collaborates with external parties to develop green products or services. GRC5: Our firm's green reputation enhances trust and long-term relationships with key partners. GRC6: Feedback from external stakeholders is regularly used to improve our environmental practices. GRC7: Our suppliers are selected and managed based on their commitment to environmental sustainability.	Wang and Juo (2021)
Sustainability culture (SUC)	A set of values, practices and beliefs to prioritise sustainable practices within the accounting profession. Seven items, were used to measure SUC, as follows: SUC1: Sustainability is a core value that guides decision-making processes within our organization. SUC2: Employees at all levels of our firm actively participate in sustainability initiatives. SUC3: Our firm prioritizes environmental and social responsibility in its daily operations. SUC4: The organization encourages innovation aimed at improving sustainability practices. SUC5: Sustainability goals are integrated into the strategic objectives of our firm. SUC6: There is clear communication and regular updates on sustainability achievements within our firm. SUC7: Management actively supports and promotes sustainability initiatives through leadership and resources.	Marshall et al. (2015); Lumpkin & Dess. (2001)
Accounting performance (ACP)	Accounting firms' performance includes: efficiency (i.e., the ability to complete accounting tasks more quickly and with optimal use of resources), accuracy (i.e., the correctness of financial data and reporting, particularly when incorporating sustainability factors into reports), innovation in reporting (i.e., the ability to adopt new and creative methods in presenting financial information, especially those related environmental sustainability and green initiatives), compliance (i.e., adherence to accounting standards, environmental regulations and sustainability disclosures), and timeliness (i.e., deliver financial reports in a timely manner, including meeting sustainability reporting deadlines). Three-item scale for each construct was used. Efficiency: [Our firm completes accounting tasks faster compared to industry standards while maintaining quality; We effectively utilize resources (e.g., time, tools, personnel) to optimize accounting processes; Our sustainability-related reporting is integrated into existing workflows without significant additional burden]. Accuracy: [Our firm's financial reports accurately reflect both financial and sustainability-related data; Sustainability disclosures in our reports are verified and free from material misstatements; Our accounting processes consistently generate correct and reliable information for all stakeholders]. Innovation in reporting: [Our firm uses innovative formats and tools to communicate sustainability performance; We frequently adopt new practices to enhance the clarity and impact of environmental reports; Our sustainability-related reporting reflects creative approaches that go beyond standard compliance]. Compliance: [Our firm consistently complies with environmental reporting standards and regulations; We regularly update our practices to align with new sustainability disclosure	Rom and Rohde (2007); DeFond and Zhang (2014); Chan and Vasarhelyi (2018); Abernathy et al. (2014); Mancini et al. (2017)

(*continued on next page*)

Table 1 (*continued*)

Construct	Conceptualisation/Operationalisation	Source
	requirements; Our financial statements and sustainability reports fully meet regulatory and industry expectations]. Timeliness: [Our firm meets deadlines for both financial and sustainability reporting; We deliver accounting reports in a timely manner to support decision-making; Our reporting timelines are consistent, even when integrating complex sustainability information].	

Source: Authors

three other accounting specialists validated its content. Five accounting academics from a Chinese university participated in the preliminary research to improve specific survey questions and statements. The research conducted a preliminary test involving 50 accounting practitioners who examined the survey questions. The research process verified both the validity and reliability of the instrument.

3.3. Data analysis tool

The researchers used SmartPLS variance-based partial least squares structural equation modeling (PLS-SEM) software for assessing the integrated model. The recommendation of Hair et al. (2017) prompted us to select PLS-SEM because the method suits non-normal data and exploratory investigation. The data collected exhibited non-normality as we obtained a p-value lower than 0.05 from the Mardia's multivariate skewness test using WebPower's online tool. The research model needed PLS-SEM as its estimation method since it proved suitable for this purpose.

The research evaluated common method bias (CMB) by studying endogenous and exogenous constructs together with the same participants through equivalent procedures (Podsakoff et al., 2003). The evaluation of CMB utilized Harmon's Single-Factor test through exploratory factor analysis according to the methodology. A study by Podsakoff et al. (2003) states that CMB does not exist when different factors emerge in analysis and the first factor only explains less than 50 % of the variance. The proposed model results showed the CMB value stood below 50 % which verified the study did not display a substantial CMB issue (Bin-Nashwan, Shah, et al., 2023).

4. Results analysis

4.1. Sample details

Accounting professionals consisting of 54 % males and 46 % females participated in the survey which demonstrates almost perfect equality between genders. The survey shows a balanced representation of data since it obtained reactions from equally split genders. Within the surveyed professional roles 48 % served as audit managers while senior auditors made up 25 % and senior accountants were 20 % with 7 % working as CFOs within the firms. In terms of professional qualifications, 40 % held CPA certifications, 35 % were ACCA members, and 19 % had CMA credentials. This variety in professional certifications contributes to a comprehensive analysis of how AI-infused knowledge-GIC influences accounting firm performance. As for educational attainment, 61 % had a bachelor's degree, 36 % held postgraduate degrees (master's or PhD), and 3 % had diplomas or other certifications. Regarding work experience, 80 % had over ten years in the field, indicating that the participants are highly experienced and well-educated, providing valuable insights into the research topic.

4.2. Measurement model

The outer model results which consist of the measurement model appear in Tables 2 and 3. The proposed model achieved satisfactory internal reliability because all latent constructs maintained values greater than 0.70 for Cronbach's α and Dijkstra-Henseler's rho and composite reliability (Table 2) (Hair et al., 2017). The assessment of convergent validity (CV) relied on checking factor loadings together with the average variance extracted (AVE). The majority of items revealed loadings surpassing 0.70 yet two items (SUC5 and ACP11) had values below 0.50 and were excluded (Table 2). The evaluation of the proposed model indicates its robust convergent validity in accordance with Hair et al. (2017).

Henseler et al. (2015) indicated that the heterotrait-monotrait ratio (HTMT) of correlation should be used for model evaluation. All the HTMT values in the study remained under the recommended threshold of 0.85 with variations between 0.540 and 0.833 according to Henseler et al. (2015). All relationships in Table 3 resulted in values less than 1.0 while the 95 % confidence intervals for HTMT values are presented in brackets below and above the actual numbers. The research confirms that the model structure allows valid and reliable measurement of concepts.

4.3. Structural model

Several preliminary standards needed to be evaluated before we can conduct our analysis of path coefficients and test the proposed hypotheses. The VIF test showed no multicollinearity since all values remained below 3.3 at the recommendation of Diamantopoulos and Siguaw (2006). We then examined the endogenous variable, accounting performance R^2 (variance explained). The analysis shows that the explanatory power of the collectively direct impact of the proposed model, i.e., GHC, GSC, and GRC, as well as the moderating role of SUC, in ADP was 42 %. On the other. This outcome indicates the significant level and good fit of the proposed model, capturing crucial drives shaping AI-infused knowledge-GIC and accounting performance. The predictive capability of the research model was further evaluated using the blindfolding technique in SmartPLS, with a predefined omission distance of seven, to calculate Stone-Geisser's Q^2 value. The findings in Table 4 indicate that the Q^2 values for the endogenous construct were scored as 0.489 above zero. This result demonstrates the satisfactory predictive relevance of the research model (Hair et al., 2017).

The proposed hypotheses were tested using PLS bootstrapping with 5000 resamples, analyzing the corresponding t-statistics and path coefficients (Chin et al., 2003). The results displayed in Table 4 reveal that GHC ($\beta = 0.221$; $p = 0.000$) and GSC $\beta = 0.225$; $p = 0.004$) within accounting firms were influenced by AI-infused knowledge (AIK) systems. Hence, we found that hypotheses H_1 and H_2 are statistically accepted. However, the impact of AIK on GRC ($\beta = 0.016$; $p = 0.865$) within accounting firms was insignificant. Thus, H_3 is rejected.

Additionally, we found that accounting firms' performance is influenced by two components of GIC, i.e., GHC ($\beta = 0.390$; $p = 0.000$), and GSC ($\beta = 0.194$; $p = 0.005$). In contrast, GRC failed to report a significant effect on accounting performance ($\beta = 0.074$; $p = 0.616$). Hence, H_4 and H_5 are accepted, while H_6 is rejected. With regard to interaction relationships, the analysis indicates that all interaction effects of GHC x sustainability culture ($\beta = 0.295$; $p = 0.001$), GSC x sustainability culture ($\beta = 0.210$; $p = 0.004$), and GRC x sustainability culture ($\beta = -0.199$; $p = 0.000$) on accounting firms' performance were significant. Thus, H_7, H_8, and H_9 are accepted.

Table 2
Outer model analysis.

Constructs	Items	Loadings	α	rho_A	CR	AVE
AI-infused knowledge	AIK1	0.882	0.821	0.820	0.895	0.736
	AIK2	0.805				
	AIK3	0.886				
	AIK4	0.798				
	AIK5	0.811				
Green human capital	GHC1	0.863	0.776	0.717	0.837	0.638
	GHC2	0.844				
	GHC3	0.828				
	CHC4	0.800				
Green structural capital	GSC1	0.889	0.721	0.715	0.873	0.768
	GSC2	0.865				
	GSC3	0.799				
	GSC4	0.822				
	GSC5	0.813				
	GSC6	0.843				
Green rational capital	GRC1	0.801	0.787	0.791	0.878	0.714
	GRC2	0.788				
	GRC3	0.838				
	GRC4	0.809				
	GRC5	0.811				
	GRC6	0.827				
	GRC7	0.851				
Sustainability culture	SUC1	0.831	0.845	0.848	0.894	0.678
	SUC2	0.839				
	SUC3	0.785				
	SUC4	0.824				
	SUC6	0.813				
	SUC7	0.816				
Accounting performance	ACP1	0.787	0.792	0.819	0.862	0.619
	ACP2	0.926				
	ACP3	0.745				
	ACP4	0.689				
	ACP5	0.861				
	ACP6	0.840				
	ACP7	0.791				
	ACP8	0.630				
	ACP9	0.823				
	ACP10	0.860				
	ACP12	0.805				
	ACP13	0.784				
	ACP15	0.748				

5. Discussion and implications

5.1. Discussion

Given the timely and relevance of AI-green intellectual capital integration and its impact on accounting professionals' performance, this study provides empirical evidence drawn from a viable model of RBV-KBV model and sustainability culture. While prior research has explored the role of AI tools in enhancing organizational performance (Mikalef et al., 2023; Olan et al., 2022; Wamba-Taguimdje et al., 2020) and the contribution of IC to competitive advantage (Ahmad et al., 2023; Kamukama & Sulait, 2017), there is a distinct lack of research examining the specific interplay between AI-infused knowledge-GIC and accounting performance in a sustainability-infused context. The literature extensively focused on traditional IC components without delving into how such elements can be adapted to the growing demand for sustainability. Furthermore, little is known about how AI impacts each dimension of GIC within accounting firms, particularly in regard to sustainability-focused innovations.

The results indicate that AI-infused knowledge (AIK) had a positive and significant impact on two components of green intellectual capital, namely, green human capital and green structural capital, while failing to affect green rational capital. It means that AI-infused knowledge systems enhance internal capabilities through the automation of tasks and the improvement of decision-making processes within accounting firms. The obtained result showing AIK positively related to GHC indicates that cutting-edge technology like AIK plays an essential role in developing the sustainability-related skills and competencies of accounting professionals. AIK platforms have the potential to facilitate personalized learning experiences, providing real-time access to sustainability resources and enhancing training programs. Such a function is more relevant to accounting firms, where professionals must stay updated on evolving sustainability-related practices and compliance regulations. By equipping accounting firms with the needful knowledge and tools, AI technologies foster accounting workforces to be better prepared to participate in green initiatives and contribute to firms' sustainability goals. While this relationship has sparsely examined in the literature, our conclusion could be relatively aligned with a prior study' results by Saxena (2024), who explored the impact of industry 4.0 on green intellectual capital.

We also found that green structural capital (GSC) is positively and significantly influenced by AI-infused knowledge. This indicates that the strong effect of IAK on accounting firms' GSC further reinforces its importance in sustainable practices. AIK provides the capability to structure operational frameworks more efficiently while advancing resource management best practices as well as data systems for higher operational effectiveness. The combination of AI-based systems enhances accounting firms' ability to address inefficiencies while conforming to environment standards and practicing sustainability more effectively through automated tasks and data analysis. Accounting firms should allocate priority to structural enhancements which will enable them to integrate sustainability-related practices across their fundamental business operations. The studies about similar findings remain

Table 4
Inner model analysis.

	Path	β	*STDEV*	*t*-value	*p*-value	Decision
$\mathbf{H_1}$	AIK - > GHC	0.221	0.061	3.622	0.000	Supported
$\mathbf{H_2}$	AIK - > GSC	0.225	0.080	2.813	0.004	Supported
$\mathbf{H_3}$	AIK - > GRC	0.016	0.095	0.169	0.865	Not supported
$\mathbf{H_4}$	GHC - > ACP	0.390	0.094	4.150	0.000	Supported
$\mathbf{H_5}$	GSC - > ACP	0.194	0.069	2.811	0.005	Supported
$\mathbf{H_6}$	GRC - > ACP	0.074	0.050	1.401	0.616	Not supported
$\mathbf{H_7}$	GHC x SUC - > ACP	0.295	0.091	3.240	0.001	Supported
$\mathbf{H_8}$	GSC x SUC - > ACP	0.210	0.073	2.876	0.004	Supported
$\mathbf{H_9}$	GRC x SUC - > ACP	0.199	0.049	4.061	0.000	Supported
		$\boldsymbol{R^2}$	$\mathbf{Q^2}$			
	ACP	0.421	0.489			

Table 3
HTMT discriminant results.

	AIK	GHC	GSC	GRC	SUC	ACP
AIK						
GHC	0.540 [0.636,0.744]					
GSC	0.742 [0.670,0.805]	0.719 [0.620,0.809]				
GRC	0.720 [0.652,0.773]	0.519 [0.420,0.600]	0.747 [0.670,0.813]			
SUC	0.753 [0.701,0.818]	0.542 [0.454,0.621]	0.737 [0.653,0.812]	0.833 [0.771,0.895]		
ACP	0.721 [0.664,0.772]	0.504 [0.415,0.585]	0.729 [0.654,0.797]	0.762 [0.702,0.815]	0.824 [0.764,0.876]	

scarce yet Goralski and Tan (2020) established that business internal practices benefit greatly from AI technologies for environmental sustainability work. The analysis discovered that AI-based knowledge created no change in green rational capital performance. Understanding the AI limitations for external relationship development becomes crucial based on these findings. Green rational capital involves networks and relationships with stakeholders like clients, venders, and the community at large, as are critical for collaborative sustainability practices. This insignificant result indicates that although AIK can streamline internal processes and individual competencies, it might not be able to effectively support interpersonal dynamics necessary to foster cooperation and trust with external stakeholders. The unsupported outcome could be attributed to the transactional nature of AI technologies, which prioritizes data processing, compliance, and financial analysis over encouraging the kind of cooperation and communication necessary to forge solid client and stakeholder relationships. Moreover, accounting clients and stakeholders may be skeptical of AI technologies (Bosse et al., 2023), viewing them as impersonal and less trustworthy than human interactions for complex financial matters. AI-infused knowledge systems may not be able to fully handle the complex and intricate stakeholder interactions in accounting that call for a thorough awareness of client-specific requirements and regulatory settings. As a result, the interplay between green relational capital and AI within accounting firms may be limited.

The study also reveals that two aspects of green intellectual capital (i.e., GHC and GSC) were positively associated with accounting firms' performance given to the various roles they play within accounting firms, while GRC was not. Green human capital—professionals' sustainability knowledge and skill—directly drives accounting firm performance through higher efficiency in sustainable practices, improved compliance with environmental regulations, and fostering innovation in financial disclosures. Despite the literature having rarely explored the GHC—accounting performance relationship, from a broad perspective prior studies revealed that firms investing in GHC can gain better performance (Mansoor et al., 2021; Yusliza et al., 2020). Likewise, green structural capital (i.e., encompasses systems, processes, and infrastructure supporting sustainability) helps to streamline operations, guaranteeing accurate and timely reporting, creative reporting techniques, and improved adherence to regulatory standards. Scholars from different contexts (e.g., Asiaei et al., 2022) also affirmed that GRC plays a substantial role in fostering firms' performance related to environmental and sustainable practices. Alternatively, we found that green rational capital—which describes firm networks and relationships with stakeholders – did not affect accounting firms' performance. This conclusion could be justified that organizational performance is infused more by internal factors, including professionals' skills related to GHC and sustainable systems related to GSC. Although green rational capital has the potential to enhance trust and long-term relationships, its effects are gradual and might not have an instant impact on the operational components of accounting firms' performance explored in our research. Studies, such as Asiaei et al. (2022), also found no effect of GRC on organizational performance.

With the introduction of sustainability culture as a moderator in our model, the analysis revealed interesting outcomes. With sustainability culture in place, all dynamics of green intellectual capital are significantly fostering accounting firms' performance. When accounting firms have a considerable sustainability culture, the value of GHC, GSC, and GRC is amplified as accounting professionals, staff, and systems are likely to be more aligned with sustainable practices. This alignment stimulates improved adherence to environmental regulations, increases productivity through the simplification of environmentally friendly processes, improves reporting accuracy through sustainable data management practices, encourages creativity in the preparation and delivery of reports, and aids in consistently meeting deadlines. We can argue that the impact of GIC on performance increases when sustainability is ingrained in the company's culture, particularly in areas where operational excellence and sustainability overlap. From the analysis, it is demonstrated that the integration of sustainability culture into the model altered the insignificant association between green rational capital and accounting performance to be significant. This result is in line with the literature (Eccles et al., 2014) showing that firms with a robust sustainability culture are more likely to engage meaningfully with stakeholders, driving improved performance outcomes.

5.2. Policy implications

The results of this research offer actionable insights for stakeholders and decision-makers, including accounting firms, clients, and the community at large. Accounting firms are highly recommended to adopt AI-infused platforms to foster GIC dynamics, specifically GHC and GSC. AI-infused knowledge tools have the potential to improve efficiency, accuracy, innovation in reporting, compliance with sustainability standards and timeliness, thus leading to better overall performance. It is suggested that firms be invested in training their staff and professionals to use AI technologies effectively, thus leveraging knowledge, skills, and competences infused within accounting firms in sustainability reporting, carbon accounting, and environmental regulations. Additionally, the impact of AI-infused knowledge on green structural capital promotes the development of advanced, AI-powered systems that support innovative methods for sustainability reporting. Firms that use AI to improve their GSC are better able to meet stakeholder expectations and regulatory requirements by providing accurate and timely sustainability reports. Accounting firms could put in place specialized training courses that cover sustainability principles and AI tools. Accounting professional skill development in utilizing AI systems for environmental reporting is facilitated by this approach, which also guarantees that green human capital is maximised for improving performance indicators such as accuracy, compliance, and innovation.

Developing a robust sustainability culture in accounting firms is essential to optimizing the influence of GIC aspects on their performance. Sustainability culture acts as a catalyst, GHC, GSC, and GRC all have a stronger impact on accounting firms' performance. Incorporating sustainability into organisational principles, plans, and accounting professionals and staff engagement initiatives is imperative for firms to optimize the advantages of AI systems and enhance compliance, efficiency, and creativity in sustainability reporting. Additionally, it is suggested that accounting firms need to pay attention to enhancing internal architecture, procedures, and systems supporting sustainability practices. This can be done by implementing AI technologies and eco-friendly platforms that facilitate process optimization, promote innovation, and minimize resources' consumption. Although green rational capital did not directly relate to performance, accounting firms could enhance their value through sustainability initiatives, building trust and collaboration with outsider stakeholders, such as clients, suppliers and regulators. Firms can align with broader social goals through strategies including involving clients in sustainability efforts and clear communication about environmental activities. These strategies can eventually translate into long-term performance gains. The study also recommends that AI solutions might be tailored to meet reporting requirements pertaining to sustainability, such as observing environmental laws. Accounting firms may enhance the speed and accuracy of sustainability disclosures and successfully address market and regulatory needs by integrating AI technology into their reporting frameworks.

While our study focuses on accounting firms, the insights gained from AIK and GIC may be applicable to sectors, such as finance, consulting, and legal services. For example, like accounting firms, these industries rely heavily on knowledge management and precision, and AI tools can enhance efficiency and sustainability efforts. However, variations in industry-specific regulations and operational structures may lead to differences in how AIK drives GIC and influences performance. Future research could explore these differences to further validate the broader relevance of AI-driven sustainability practices across various

service sectors.

5.3. Theoretical implications

Upon a thorough review of the literature, this research is a pioneer and provides profound implications to the theory and theoretical framework of the interplay between AI-infused knowledge and green intellectual capital, and eventually how to shape accounting firms' performance. The contributions of this study are summarized as follows: First, the literature has mainly paid attention to the traditional aspects of intellectual capital and firm performance (Bayraktaroglu et al., 2019; Truong et al., 2024; Xu & Li, 2022), green intellectual capital and firm performance (Hoang Thanh & Truong Cong, 2024; Shehzad et al., 2023; Umar et al., 2024), and the role of intellectual capital in firms' competitive advantage (Ahmad et al., 2023; Kamukama & Sulait, 2017). However, there is a dearth of research exploring the crucial impact of AI-infused knowledge on green intellectual capital, and thus fostering firms' performance. Therefore, the present study is likely the first of its kind to delve empirically into the interplay between AI-infused knowledge on green intellectual capital. Second, the green intellectual capital—firm performance relationship was examined in the literature to a reasonable extent, it is not overly saturated compared to traditional intellectual capital. Prior studies have investigated the effect of GIC on performance in certain domains, such as manufacturing (Umar et al., 2024; Shehzad al., 2023; Yusliza et al., 2020; Mansoor et al., 2019), hospitality (Elzek et al., 2024; Haldorai et al., 2022; Shahbaz et al., 2024), high-tech industry (Wang & Juo, 2021), SMEs (Khanlarov et al., 2020, pp. 100–109), healthcare (Zahlan et al., 2023), energy (Liu et al., 2021), education (Ahmed et al., 2023), and waste recycling (Haddad et al., 2024). Yet, although service-oriented contexts, like accounting, also have the potential to benefit from green intellectual capital practices, research in such a significant discipline has been remarkably scarce. Green strategies adopted in accounting firms could spur environmental reporting, improving stakeholder engagement around sustainability and optimizing internal processes to be eco-friendlier. Hence, our study enriches the literature by examining the interplay of AI-infused knowledge and green intellectual capital within accounting firms and how this can spur their performance. Third, our study also contributes to the body of knowledge by incorporating sustainability culture as a moderator, strengthening the association between green intellectual capital and accounting firm performance. Fourthly, the integrated model of RBV-KBV could provide a viable and comprehensive framework for AI-infused green intellectual capital and firm performance. This model could serve as a catalyst for further literature in the field.

6. Conclusions, recommendations and future directions

Although a growing body of literature has been drawn to the nexus between green intellectual capital (GIC) and firm performance, little is known about how AI technologies can enhance green intellectual capital and eventually firm performance. This study aims to explore the impact of AI-infused knowledge on GIC within accounting firms, and thus potentially fostering their performance. Relying on a quantitative research approach employed to collect data from accounting firms, the analysis results show that AI-infused knowledge had a significant impact on the internal two aspects of GIC, green human capital (GHC) and structural capital (GSC), while it had no impact on green rational capital (GRC). Internal dimensions, i.e., GHC and GSC, had a significant impact on accounting performance across various aspects, including efficiency, accuracy, compliance with environmental standards, innovation in reporting, and timeliness. Sustainability culture was found to significantly moderate the association between GIC dimensions and accounting firm performance.

Drawn from our results, the study suggests some actionable recommendations to stakeholders: With the integration of AI-infused knowledge, accounting firms are recommended to pay attention to investment in extensive training programs focusing on sustainability practices and green accounting principles. Integrating AI-enhanced knowledge frameworks can substantially foster GHC by giving staff and professionals access to cutting-edge resources and real-time data that can infuse their understanding of sustainability practices. With AI-powered insights adoption, firms can efficiently upskill their employees in green accounting concepts, enabling them to integrate sustainable initiatives into their business model. Thus, this can lead to innovative sustainability-related accounting activities, enhancing the firm's overall performance. Additionally, efforts must be geared toward the vital role of AI-infused knowledge in fostering GSC by automating sustainability processes, enhancing reporting accuracy, and embedding sustainability metrics into accounting systems. AI solutions, for example, have the potential to make environmental data collection and analysis more efficient and accurate, allowing accounting firms to embrace more effective and precise sustainability reporting procedures. By integrating AI-fueled knowledge into green structures and processes within accounting firms, robust sustainability structures and policies can be built, and green initiatives can be consistently integrated. Effective analysis of vast datasets using AI technologies can facilitate long-term planning and strategic decision-making in line with sustainability goals. Another area that warrants further consideration is green rational capital and how accounting firms can employ AI-infused knowledge systems to improve their stakeholder engagement. This could be done by transparent, data-driven communication about sustainability. AI solutions can assist these companies advance trust and collaborate with stakeholders more effectively. It is suggested that companies that have a robust sustainability culture backed by AI-driven insights are better able to meet stakeholder needs and match their green initiatives with market expectations. By leveraging AI capabilities in amplifying sustainability activities, firms have the potential to demonstrate strong sustainability practices and bolster their rational capital. This, in turn, can lead to improved compliance, efficiency, and innovation in accounting firm performance.

However, despite the promising implications of the study, some limitations are to be underlined. The study employed a cross-sectional survey, capturing the nexuses between AI and GIC, and accounting firm performance at a single point in time. Given the limitation of this approach in assessing long-term causal effects, longitudinal future research could provide deeper insights into the evolution of such a model. Moreover, as accounting firms operate in specific regulatory and market environments, this is likely to influence AI adoption and sustainability practices. Accordingly, some crucial extrinsic factors, including market competition, regulatory pressures, and client expectations, which may have an impact on the development of green intellectual capital and performance were not incorporated into the model. Finally, our study relied on the service-oriented sector (accounting firms), as this may restrict the generalization of the outcomes to other sectors. It is suggested to expand the sample in future research by including diverse industries.

CRediT authorship contribution statement

Saeed Awadh Bin-Nashwan: Writing – review & editing, Writing – original draft, Resources, Project administration, Methodology, Formal analysis, Conceptualization. **Jackie Zhanbiao Li:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization.

Data availability

Data will be made available on request.

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